

GATE Question Paper 2010, EE Question Number 52

Question 12 Analysis

Question:

The following Karnaugh map represents a function F .

		00	01	11	10
X	0	1	1	1	0
	1	0	0	1	0

52. A minimized form of the function F is

- (A) $F = \bar{X}Y + YZ$ (B) $F = \bar{X}\bar{Y} + YZ$ (C) $F = \bar{X}\bar{Y} + Y\bar{Z}$ (D) $F = \bar{X}\bar{Y} + \bar{Y}Z$

4. Final Expression:

$$F = \bar{X}\bar{Y} + YZ$$

Implementation using Pico W

Abstract

This part replicates the same logic on Raspberry Pi Pico W using MicroPython via Thonny IDE.

Hardware Requirements

S.No	Component
1	Raspberry Pi Pico W
2	Breadboard
3	Push Buttons (X, Y, Z)
4	LED
5	Resistors: 220 Ω , 10k Ω
6	Jumper Wires
7	Micro USB Cable

Table 1: Hardware Components

Karnaugh Map Simplification Problem

Given: The following Karnaugh map represents a function F :

Step-by-Step Solution

1. Identify variables:

- Variables: X, Y, Z
- Columns in Gray code: 00, 01, 11, 10

2. Extract minterms:

- 1s at: $(0, 00), (0, 01), (0, 11), (1, 01)$
- Minterms:

$$(0, 00) \rightarrow \bar{X}\bar{Y}\bar{Z}$$

$$(0, 01) \rightarrow \bar{X}\bar{Y}Z$$

$$(0, 11) \rightarrow \bar{X}YZ$$

$$(1, 01) \rightarrow X\bar{Y}Z$$

3. Group minterms:

- Group 1: $\bar{X}\bar{Y}$
- Group 2: $\bar{X}YZ + X\bar{Y}Z = YZ$

Pin Connections

Component	Pico W Pin
X	GP14
Y	GP15
Z	GP16
F (LED)	GP13
GND	GND
VCC	3.3V

Table 2: GPIO Pin Config

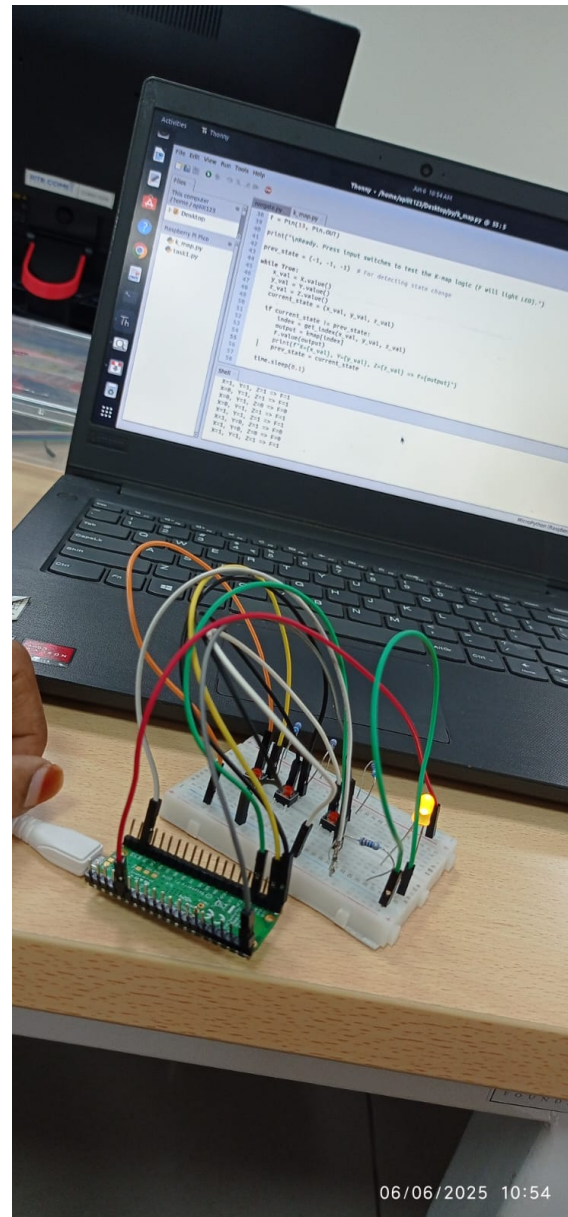
Upload Steps

1. Connect Pico W while holding BOOTSEL
2. Copy .uf2 firmware to RPI-RP2
3. Open Thonny IDE and set interpreter to Pico
4. Write and run MicroPython code
5. Connect hardware circuit and test

Observations

X	Y	Z	F (LED)
0	0	0	1
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0
1	1	1	1

Table 4: Truth Table



https://github.com/madhuu34/Madhulathafwc/blob/main/Hardware/PLATFORM_I_O/hardware_code.py

Conclusion

The logic function was successfully tested on Pico W using MicroPython and validated with real input/output.